New Horizons for Multiple Sclerosis Therapy: 2025 and Beyond

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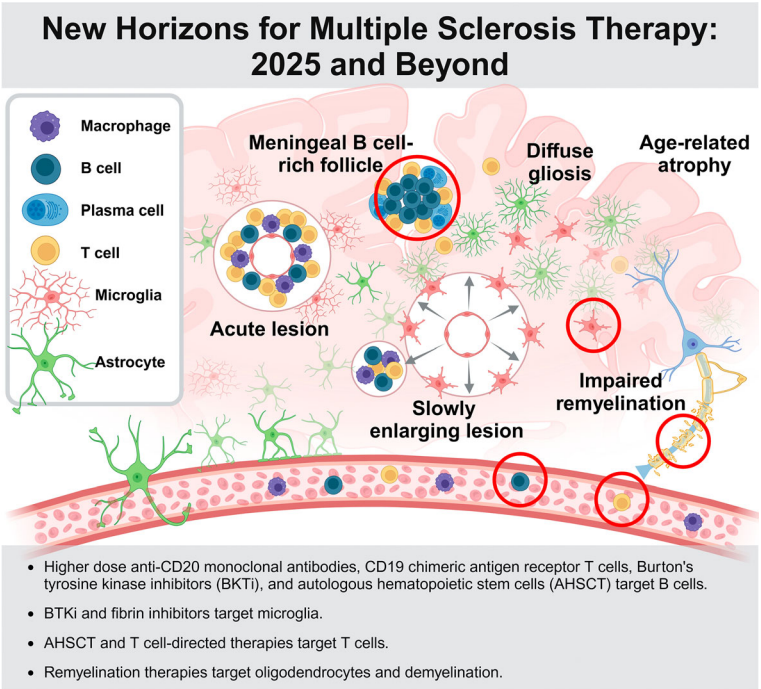

The advances achieved against multiple sclerosis (MS) represent one of the great success stories of modern molecular medicine. The development of therapies with increasing selectivity and safety, guided by gains in understanding the fundamental immunology, neurobiology, genetics, and triggers of this disease, have broadened the traditional focus on treatment, adding realistic possibilities for prevention and repair. Here, we summarize recent advances that have together transformed the disease from the most common crippler of young adults in the western world to its place today as a condition in which most newly diagnosed patients can anticipate lives free from disability.

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The Current Therapeutic Landscape

Monoclonal antibody (mAb) disease-modifying therapies (DMTs) that target CD20 (rituximab, ocrelizumab [OCR], ofatumumab [OFA], and ublituximab), CD52 (alemtuzumab), and α4 integrin (natalizumab) are highly effective at relapse prevention and blocking new brain lesion formation. A growing body of long-term outcome data supports their early use.[1–7] These DMTs either deplete[8–11] or block migration[12,13] of white blood cells into the central nervous system (CNS). The anti-CD20 therapies, which are the most selective of the high-efficacy options, have emerged as the first-line therapy of choice for many patients worldwide. These agents exclusively target mature B lymphocytes, and they spare early B cell precursors, late plasmablasts and plasma cells, and other immune cells (a small proportion [~6%] of T lymphocytes also express CD20; they are functionally heterogeneous and their role, if any, in MS is uncertain). This high degree of selectivity likely underlies the excellent safety record of the anti-CD20 therapies with chronic use. In a larger sense, the mechanism of action of all of these therapies implies that acute CNS inflammation and demyelination in multiple sclerosis (MS) is mediated by the influx of pathogenic lymphocytes across the blood brain barrier (BBB), a process sometimes referred to as "outside-in" pathology.[14] The impact of DMTs on lesion formation provides the strongest evidence to date for direct involvement of the peripheral immune system in relapsing MS disease activity.

Despite the acute relapse being the most readily recognized clinical hallmark of MS, relapse biology remains relatively poorly understood. What directs the occurrence, location, or frequency of MS relapses is unknown. Further, the determinants of lesion biology, including their formation and evolution into either chronic active lesions or inactive lesions, are poorly understood. The extent of axonal injury incurred during lesion formation is presumed to determine the degree of recovery, although other factors including neuronal plasticity, capacity to restore conduction across injured axons, and remyelination all likely contribute to post-relapse recovery.[15]

The success of B-lymphocyte targeting anti-CD20 mAbs in treating relapsing MS indicated that B cells play a more central role in MS pathogenesis than previously thought. Pluripotent B cells exert a variety of effector functions that can modulate not only the humoral immune system, but cell-mediated and innate immunity as well. The presence of oligoclonal bands (OCBs) are a diagnostic hallmark of MS[16–18] and implies that dysregulation of CNS resident cells from the B cell lineage are involved in MS pathogenesis, although no definitive disease-specific antigenic targets have been found.[19] In MS patients, memory B cells produce higher than normal levels of pro-inflammatory cytokines, including interleukin-6 (IL-6), granulocyte-macrophage colony-stimulating factor (GM-CSF), and tumor necrosis factor-α (TNFα).[20–23] The therapeutic effects of anti-CD20 mAbs are probably unrelated to changes in antibody production, because total immunoglobulin G (IgG) levels generally remain unchanged over many years of treatment,[24,25] and OCBs persist despite peripheral B cell depletion.[26,27] Moreover, the targets recognized by CNS-produced antibodies, appear to be diverse rather than related to a common antigen.[28] This observation argues against MS being caused by a pathogenic antibody against a single antigenic target, as is the case for neuromyelitis optica spectrum disorder (NMOSD)[29] and many forms of autoimmune encephalitis (AE).[30] However, it is still possible that the earliest MS event might consist of a highly focused autoimmune response to a triggering stimulus that then spreads over time to involve a diverse group of antigenic targets.

Increasing evidence points toward the importance of antigen presentation by B cells to T cells in MS pathogenesis. B cells express major histocompatibility complex II (MHC II) molecules (as well as MHC I, which is expressed by all nucleated cells), allowing them to directly activate T cells. B cells in the CSF and CNS of MS patients express increased levels of MHC proteins and co-stimulatory molecules[31,32] and can activate autoreactive T cells in MS and animal models.[33–36] One particularly intriguing finding is that B cells from MS patients have a propensity to present digested fragments of their own MHC molecules to T cells, stimulating responding T cells that cross-react with MS-relevant antigens, including epitopes of myelin, Epstein–Barr virus (EBV), and Akkermansia mucinophilia bacillus.[37] B cells also receive feedback from T cells, in particular from T follicular helper cells, which are themselves increased in MS patients.[38,39] Therefore, the B cell-T cell axis is believed to be a critical linchpin in pathophysiology. Disruption of this link is hypothesized to underlie the therapeutic effects mediated by different DMTs.

Despite the impact of anti-CD20 mAbs in preventing relapses and new lesions, disability worsening in progressive forms of MS (PMS) is only modestly reduced.[40–42] Why PMS responds incompletely to treatments that are highly effective in relapsing forms of MS (RMS) is unknown. One hypothesis is that compartmentalized inflammation within the CNS is refractory to treatment with mAbs that are poorly penetrant across the BBB.[43] However, small molecule DMTs that can cross the BBB are also relatively ineffective in reducing disability accumulation in PMS.[44–46] Taken together, the limited success of peripherally administered mAbs and failure of brain penetrant small molecule anti-metabolites suggests that pathological

processes other than dysregulated adaptive immunity also contribute to PMS.

The Challenge of Progression

Tissue damage in MS results from bouts of peripherally mediated inflammation (new focal lesion formation) as well as other, more insidiously progressive pathologies. Irreversible disability is believed to result predominantly from cumulative neuronal loss. Gray matter atrophy can be present at symptom onset, suggesting that the disease process begins very early. Further, this process causes diffuse, or perhaps even global, tissue loss.[47]

The histopathologic and radiographic basis for progression remains poorly delineated. Correlations of acute lesions or overall T2 burden of disease with progression are poor (the so-called clinical-radiologic paradox).[48] Irreversible tissue injury causing loss of brain and spinal cord volume over time accompanies progressive disability, yet none of these features are specific for progression. PMS shares similarities with neurodegenerative diseases in that atrophy accompanies insidious loss of function. Progressive disability is now recognized to occur in many patients much earlier in the course of the disease than was previously recognized, a finding recognized by long-term studies of RMS cohorts whose disease activity (both clinical relapses and new MRI lesions) was effectively suppressed by DMTs.[49] In contemporary datasets, most confirmed disability events in RMS occur without any prior relapse, a phenomenon termed silent progression in RMS or progression independent of relapsing activity (PIRA) across the disease spectrum.[50,51] Because these events can occur early in the course of the disease and in patients with few or no impairments on neurologic exam, it is clear that progressive disability is not exclusively a secondary or late-stage process, but rather it occurs across the continuum of MS. These observations also indicate that pathological processes distinct from those that control new lesion formation are likely to underlie progressive biology.

Pathologic features associated with PMS include widespread cortical demyelination with neuronal loss, chronic active plaques with microglial edges, prominent axonopathy, diffuse white matter injury with astrogliosis, perivascular fibrin deposition, mitochondrial dysfunction, and global brain and spinal cord atrophy.[52–56] However, none of these features are specific for progression: all can be present early in the course of the disease and their accumulation over time may simply reflect the consequences of chronic neuroinflammation rather than a unique pathology underlying progressive disability. Cortical demyelination occurs in close apposition with meningeal immune follicle-like structures consisting of B cells and T cells.[47,48] CD20- CD138+ plasma cells are also present in the leptomeninges in PMS, as well as in parenchymal white matter.[57,58] To what extent these CNS resident plasma cells contribute to MS pathogenesis is not known. Nevertheless, plasma cells could contribute to intrathecal gammaglobulin production. In addition, cytotoxic CD8+ T cells are particularly conspicuous at leading edges of tissue damage.[59]

Although T cells and B cells persist in MS lesions, they are found in lower numbers in progressive MS lesions compared to acutely inflamed lesions,[60] possibly indicating a reduced role of the adaptive immune system later in the disease course. Conversely, the innate immune system appears to play a greater role in progressive MS. Microglia, in particular, demonstrate widespread activation in progressive MS, including throughout the normal appearing white and gray matter.[54] Activated microglia are associated with axonal injury, neuronal loss, and synaptic pruning.[61] T cells are found in close proximity to activated microglia in normal appearing white and gray matter,[62] suggesting potential T cell activation by microglia. Chronically activated microglia appear to also induce a neurotoxic astroglial phenotype that could in turn cause neuronal injury.[63]

Microglial nodules can be found in the normal-appearing white matter of patients with MS where they appear to respond to antigen presentation and demonstrate phagolysosomal activity.[64] These observations suggest a role for microglia in diffuse white matter injury in MS or possibly early lesion formation. Microglia proliferate in acute active MS lesions and as the lesions evolve, collect predominantly at the lesion border rather than the lesion center.[65] In chronic active lesions, there is gradual concentric expansion with microglia at the leading edge plus progressive tissue damage within. As lesions become inactive, microglial activation subsides. One interpretation of these histopathological observations implicates microglia as mediating tissue injury, however, it is also possible that some microglia function to limit damage by mediating debris clearance.[66] Even within lesions, there appear to be different populations of microglia based on expression profiling. Foamy microglia appear to be involved in phagocytosis with upregulated lipid processing and lysosomal activity, whereas iron-laden microglia have upregulated gene expression profiles for antigen processing, complement activation and iron processing.[67] In murine models of CNS inflammation such as experimental autoimmune encephalomyelitis (EAE) as well as the cuprizone model of toxic demyelination, microglial depletion confers a protective role,[66] indicating that in these models microglia appear to participate as effectors of tissue injury.

Activated microglia play a critical role in sensing pathogens and responding to tissue injury. Secretion of TNFα, IL-1, and complement component 1q (C1q) promotes a neurotoxic astroglial cell phenotype that produces reactive oxygen species and saturated lipids, which in turn damage

neurons and oligodendroglia.[68] Other mechanisms of injury that might contribute to MS pathology include inhibition of oligodendrocyte precursor cell proliferation via IL-1 production, enhanced synaptic remodeling and potentiation of macrophage inflammation.

Finally, although remyelination occurs to varying degrees in demyelinated lesions, it is greatly reduced in progressive MS lesions[69] and with aging. This suggests that therapeutic targets for treating progressive MS might target glial cell function to not only reduce neuroaxonal injury, but also to promote myelin repair.

Moving beyond Suppression of Lesion Formation and Relapses

Near-complete suppression of focal inflammatory disease activity and new magnetic resonance imaging (MRI) lesions is now achievable in the great majority of relapsing patients treated continuously with high-efficacy therapy. In the ongoing extension trials of anti-CD20 B cell therapy with OCR and OFA, patients experience on average fewer than one relapse each 20 to 30 years, a complete cessation of new brain MRI lesions, and reductions of serum neurofilament light chain (NfL) levels to healthy normal levels.[70] These findings, combined with real-world data, indicate that serial MRI scans may no longer be needed in clinically stable patients receiving these therapies. Yet as noted above, it is also clear that progressive disease activity, while attenuated, persists in most patients, and is not adequately captured with current imaging and protein biomarker measures. Patients receiving anti-CD20 therapy achieve year-to-year whole brain volume reductions that approximate age-matched healthy controls (~0.4% annually) despite experiencing ongoing silent progression. Atrophy progression in the high cervical spinal cord has shown promise as a more sensitive correlate of progression,[71] but has not yet been applied to formal prospective trials of therapy. The development of increasingly sensitive serum protein biomarkers, such as glial fibrillary acidic protein (GFAP), also raise hope that in the future these may provide clinically useful markers of insidious progression.[72] Positron emission tomography-based imaging using the 18-kDa translocator protein (TSPO) can also provide insight into microglial activation,[73] which may be predictive of disability progression.[74,75] However, it is important to recognize that TSPO signal can reflect a variety of CNS cellular sources,[76] therefore, more selective microglia markers are solely needed.

The Next Era of Therapeutics

Next Generation B Cell Depletion Strategies

Despite near complete control of lesion formation and relapses, the impact of B cell depletion therapy on confirmed disability progression (CDP) is approximately 40% in RMS and 30% (with OCR) in primary progressive multiple sclerosis (PPMS).[77] More complete control is sorely needed, and one possibility is that higher dosing might achieve superior results. In the OCR pivotal studies, post hoc analyses indicated that RMS patients who received higher effective doses of IV OCR experienced less silent progression, although the added benefit in PPMS was less apparent,[78] and preliminary results from a large prospective trial (MUSETTE) in RMS were also negative. Furthermore, a similar analysis in RMS with SQ OFA did not show any dose–response relationship with respect to progression.[79] Therefore, data to date indicate that the benefits of anti-CD20 therapies against progression are not improved with higher dosing regimens, and alternative strategies will be required.

Targeting Plasmablasts and Plasma Cells

In chronic MS, B cell inflammatory infiltrates in the CNS are comprised predominantly of differentiated, immunoglobulin (Ig)-secreting plasmablasts and plasma cells.[80] Importantly, the CD20 surface antigen is not expressed on these late-stage B cells, therefore rendering them unaffected by anti-CD20 therapy. Although it is likely that over time these cells could be gradually depleted because they are no longer replenished by CD20+ precursors, some plasma cell clones can persist for a lifetime. Therapies that target these CD20-negative B cell populations are, therefore, attractive candidates to provide more complete B cell depletion. The CD19 surface antigen is especially promising, as it is present not only on early and mature B cells, but also on plasmablasts. Although there are several anti-CD19 monoclonal antibodies approved for use in the United States, and one, inebilizumab, is approved for treatment of neuromyelitis optica,[81] these have not been rigorously tested in MS. Preliminary trials suggested that inebilizumab was effective in RMS,[82] but whether or not there is any benefit against progression is unknown.

A CD19-targeted approach that has attracted considerable recent attention uses chimeric antigen receptor T (CAR T) cells for the treatment of autoimmunity. CAR T cells are typically autologous T cells, harvested by lymphocytopheresis, that are genetically engineered to target a specific antigen and have revolutionized treatment of several malignancies.[83] Case series in systemic lupus erythematosus (SLE), and in other autoimmune diseases such as myositis, have indicated that depletion of B cells with CAR T cells can be highly effective.[84] In SLE, a single infusion of CD19 CAR T cells led to dramatic reductions in clinical symptoms and serologic autoantibody biomarkers of active disease. Moreover, these benefits persisted long after B cell repletion, which generally occurred 3 months after treatment, suggesting that the

pathogenic immune response had been at least partly ablated. These results are even more remarkable when measured against the failure of pivotal trials testing anti-CD20 therapies in SLE. If the CAR T cell experience is confirmed in larger studies, this would suggest that plasmablasts are a key culprit in lupus or alternatively that CAR T therapies provide more effective depletion of B cells in extravascular immune niches than do monoclonal antibodies. Although it is possible that the chemoablative "conditioning" regimens required before infusion of CAR T cells could be responsible for some of the observed benefits, this is unlikely given the very modest benefits observed in earlier studies of generalized immune suppression for SLE. To date, only a few MS patients treated with CD19 CAR T cells have been reported,[85] but this is an area of investigation that is certain to rapidly expand (Table). It seems likely that CAR T cell therapies will be as effective as anti-CD20 monoclonal antibodies for treatment of MS, but whether they will provide more durable responses and greater benefits against progression, and if these advantages are worth the cost, risks, and complexities of treatment will need to be defined.

"Rebooting" the Immune System

The CD19 CAR T cell experience suggested that this approach might provide long-lasting, or even permanent, elimination of B cell clones producing pathogenic autoantibodies that drive some autoimmune diseases. For MS, however, as noted earlier no culprit autoantibodies have been identified. CSF OCBs could represent a relevant indicator of persistent autoimmunity, with the caveat that the targets of most OCBs appear to be heterogeneous and are not directed against defined brain antigens.[19]

CAR T cells directed against B cell maturation antigen (BCMA), which is highly expressed in plasma cells, have recently been shown to be effective against B cell malignancies such as multiple myeloma.[86] Using a high throughput screening platform capable of assessing the full repertoire of autoantibodies directed against linear self-peptides (termed the autoreactome) has revealed that the autoreactome is rebooted in patients treated with BCMA CAR T cells, but not CD19 CAR T cells.[87] These data raise hope that BCMA-targeted therapies may offer a long-lasting immune reset for B cell mediated autoimmune conditions including MS. However, adverse events including parkinsonism and cranial neuropathies were reported in some patients treated with BCMA CAR T cells. BCMA is expressed at low levels in the CNS, and this safety signal may represent an adverse on-target effect.[88]

With advances in cellular engineering, a new generation of cell therapies is emerging that will permit selective targeting to any antigen or organ of interest, and to deliver therapeutic payloads, such as IL-10, to downregulate a proinflammatory environment.[89] Moreover, viral vector-based gene delivery approaches could one day be used to directly engineer CAR T cells of any specificity in vivo, obviating the need for cell harvesting, gene insertion, and reinfusion.[90]

Autologous hematopoietic stem cell transplantation (AHSCT) represents another approach to rebooting in autoimmune diseases, with effectiveness reported in treatment-resistant conditions such as scleroderma.[91] Immunoablative conditioning is also required for AHSCT before bone marrow harvesting and re-infusion of expanded CD34-positive stem cells, and the immunosuppressive regimens used have varied from relatively mild to severe. When RMS patients are treated early in their disease course, case series have reported highly favorable moderate-term (eg, 5 years) outcomes, with at least 75% of patients achieving stabilization of clinical and radiologic outcomes, and improvement of clinical disability has also been noted.[92] Most reports have been retrospective. In one series that used a propensity score method to match patients with those receiving high-efficacy DMTs, AHSCT-treated patients had outcomes superior to natalizumab but similar to OCR.[93] Patients with established primary or secondary PMS do not appear to respond convincingly.[92] Whether the durable responses reported after AHSCT reflect suppression of disease associated with an increase in systemic immune regulatory tone[93] or a permanent reboot is uncertain. Some studies report disappearance of OCB over time,[94] but others highlight a re-emergence of myelin-reactive T cells.[93] Multiple prospective trials are currently underway that are expected to provide a more complete picture of the value, risk, and immune effects of AHSCT in RMS (Table).

Beyond B Cells

As noted above, other immune cell types, including activated microglia and CD8+ T cells, are also associated with ongoing tissue injury in chronic MS. Attention has focused on microglia, the major cell type mediating innate immunity in the CNS, not only in MS but other neurodegenerative conditions as well. An unmet challenge is the considerable heterogeneity of microglial cell types that exist in the human CNS both in health and disease, some with contrasting effects on immunity, and uncertainty with respect to which microglia are culprits. An ideal therapeutic would target only the responsible microglial subtype, a goal that is not currently possible.

Microglia and Bruton's Tyrosine Kinase Inhibitors

Bruton's tyrosine kinase (BTK) was first identified as the gene associated with X-linked agammaglobulinemia (XLA or Bruton's agammaglobulinemia).[95] BTK is a tyrosine

TABLE. Select Therapeutic Strategies Currently in Development

Compound	Phase	NCT no.	Sponsor	Study population	N	Results
CAR T cellular therapies						
KYV-101 (CD19)	1	NCT06138132	Stanford University	PMS	12 (est)	Ongoing
	1	NCT006451159	UCSF	PMS	10 (est)	Ongoing
	2	NCT06384976	Kyverna	PMS	120 (est)	Ongoing
CC-97540 (CD19)	1	NCT06220201	Juno Therapeutics, Bristol-Myers Squibb	MS	98 (est)	Ongoing
YTB323 (CD19)	1/2	NCT06617793	Novartis	RMS	28 (est)	Ongoing
Stem cell therapies						
AHSCT vs alemtuzumab	3	NCT03477500	Haukeland University Hospital	RMS	100 (est)	Ongoing
AHSCT vs best available therapy	3	NCT04047628	National Institute of Allergy and Infectious Diseases (NIAID)	RMS	156 (est)	Ongoing
Adipose derived MSC	2	NCT05116540	Hope Biosciences Stem Cell Research Foundation	RMS	24	Positive
Bruton's tyrosine kinase inhibitors						
Tolebrutinib	3	NCT04410978	Sanofi	RMS	974	Negative
	3	NCT04410991		RMS	899	Negative
	3	NCT04411641		nrSPMS	1,131	Positive
	3	NCT04458051		PPMS	767	Ongoing
Fenebrutinib	2	NCT05119569	Genentech/Roche	RMS	109	Positive
	3	NCT04586023		RMS	751	Ongoing
	3	NCT04586010		RMS	746	Ongoing
	3	NCT04544449		PPMS	985	Ongoing
Remibrutinib	3	NCT05156281	Novartis	RMS	800 (est)	Ongoing
	3	NCT05147220			800 (est)	Ongoing
Orelabrutinib	2	NCT04711148	Beijing InnoCare Pharma Tech	RMS	160	Positive
BIIB091	2	NCT05798520	Biogen	RMS	275 (est)	Ongoing
Remyelinating therapies						
Clemastine fumarate	2	NCT02521311	UCSF	AON in RMS	90 (est)	Ongoing
Clemastine fumarate + metformin	2	NCT05131828	Cambridge University Hospitals NHS Foundation Trust	RMS	70 (est)	Ongoing
Bazedoxifene acetate	2	NCT04002934	UCSF	Post-menopausal women with RMS	62 (est)	Ongoing
PIPE-307	2	NCT06083753	Contineum Therapeutics/Johnson & Johnson	RMS with VEP latency delay	168	Ongoing
PIPE-791	2	NCT06683612	Contineum Therapeutics	PMS, HC, IPF	28 (est)	Ongoing

AHSCT = autologous hematopoietic stem cell; AON = acute optic neuritis; CAR T = chimeric antigen receptor T Cells; est = estimated recruitment target; HC = healthy control; IPF = idiopathic pulmonary fibrosis; MSC = mesenchymal stem cell; nrSPMS = non-relapsing secondary progressive multiple sclerosis; PMS = progressive forms of MS; PPMS = primary progressive multiple sclerosis; RMS = relapsing forms of MS; VEP = visual evoked potential.

kinase essential for transmitting signals from the pre-B cell receptor that forms following Ig heavy chain rearrangement. In addition, BTK can signal through toll-like receptors. In the absence of BTK, pro-B cells fail to mature into pre-B cells and do not produce antibodies. Affected children have lymphoid hypoplasia and low levels of all classes of Igs. In patients with XLA, pre-B cells fail to mature and enter the circulation. This rare hereditary condition manifests in male children who present with recurrent infections of the ears, sinuses, lungs, and skin.[96]

BTK is involved in proliferation in B cell cancers and, therefore, is a target for treatment of malignancies such as mantle cell lymphoma and chronic lymphocytic leukemia. Four BTK inhibitors (BTKi) are United States Food and Drug Administration approved for these indications: ibrutinib, acalabrutinib, zanubrutinib, and pirtobrutinib. These drugs have high response rates with long durations of efficacy, however, these drugs cross-react with other kinases and have off-target effects. Further, because of poor CNS penetrance they have limited application to CNS B cell malignancies. Nonetheless, the successful use of these products in long-term treatment of chronic B cell malignancies demonstrates the potential for successful application of BTKi to other B cell-mediated diseases.

In addition to B cells, BTK is present in microglia and macrophages, and also in mast cells where it plays a role in activation following IgE receptor engagement. Inhibiting BTK activity reduces microglial inflammatory responses and BTK inhibition is effective in EAE. BTK inhibition could, therefore, have multiple beneficial effects in MS mediated by effects on each of these cell types. Although both B cells and macrophages could be inhibited in the periphery, targeting microglial BTK requires a CNS-penetrant BTKi.

Several BTKi are in later development for MS (Table). The first to complete its phase 3 clinical trial program in relapsing MS was evobrutinib that failed to show superiority to teriflunomide on any outcomes despite encouraging results in a phase 2 trial, which compared evobrutinib to dimethyl fumarate or placebo.[97] Insufficient CNS penetrance for robust receptor occupancy at the dose used in the trial was proposed as an explanation for this program's failure. Similar to evobrutinib, tolebrutinib also failed to show superiority to teriflunomide on reducing relapse risk in its phase 3 trial. However, tolebrutinib outperformed teriflunomide on reducing the risk of disability worsening, although this effect was considered nominal because of the failure of success on the primary endpoint.[98] Tolebrutinib was also studied in trials of non-relapsing secondary progressive MS (nrSPMS) and PPMS. The trial in nrSPMS succeeded in meeting the primary endpoint on disability prevention and showed a favorable impact on disability improvement.[99] As such, tolebrutinib is the first drug with proved efficacy in this group of patients for whom there is a significant unmet need. A trial of tolebrutinib in PPMS is ongoing at present and is hoped to yield similar results given the impact of tolebrutinib in nrSPMS. In contrast to evobrutinib whose CNS penetrance was limited, tolebrutinib was developed to be highly CNS penetrant and this difference in bioavailability could account for the differential impact of the two products on disability. If this hypothesis is correct, then inhibition of BTK in microglial cells, or other BTK-expressing cells within the CNS, may reduce MS-related disability beyond the modest impact on relapse rate reduction. Two other BTKi, fenebrutinib and remibrutinib, are in late development. In contrast to evobrutinib and tolebrutinib, these agents have the advantage of higher selectivity for BTK with less cross-activity with other kinases. Fenebrutinib and remibrutinib are being compared to teriflunomide in relapsing MS, and fenebrutinib is also being studied in PPMS in a head-to-head trial versus OCR. The fenebrutinib trials are estimated to complete in early 2026, and the remibrutinib studies in relapsing MS remain open for recruitment. The results of these trials are likely to define the value of a BTKi approach for MS.

Targeting pro-Inflammatory Activators of Microglia

Another promising approach is to interfere with fibrin, an activator of pro-inflammatory microglia.[100] Fibrin, which is widely deposited in the subendothelial spaces in MS, triggers microglial activation and consequent injury to neurons and myelin through recognition of an epitope distinct from that which activates the clotting cascade. Monoclonal antibodies have been developed that block this immunostimulatory region of fibrin, leaving its clotting function intact, and these represent one promising approach to counter microglial-induced injury.

Cytotoxic CD8+ T Cells

CD8+ T cells are the most abundant lymphocyte population in MS lesions where they are present in the perivascular space as well as lesion parencyhma.[60] They also demonstrate high clonal expansion,[101] suggesting encounter with local CNS antigen. CD8+ T cell phenotyping in MS lesions and normal appearing white matter indicates a chronically activated, cytotoxic profile. There are currently no specific therapies targeting CD8+ T cells in autoimmunity, however, such a therapeutic approach could represent a promising new direction for these patients. Although long-term depletion of CD8+ T cells would likely carry significant risks (ie, infection, malignancy), treatments targeting a subset of

pathogenic CD8+ T cells or with short-term depletion (as occurs with alemtuzumab) may provide safer strategies.

Earlier Treatment

A large body of evidence from both experimental models of autoimmunity and human autoimmune diseases indicates that opportunities to eliminate a pathogenic autoimmune response are greatest when treatment is initiated at the earliest possible time. In autoimmune diseases such as SLE, insulin-dependent diabetes mellitus (IDDM) and rheumatoid arthritis (RA), the presence of serologic autoantibodies have revealed that there is a hierarchical appearance of autoantibodies that develop during the presymptomatic stage, with "benign" early appearing antibodies appearing first, and more pathogenic ones associated with tissue damage developing later.[102] These human data also validate a large body of evidence in models that autoimmunity can begin as a highly focused response, perhaps to a viral or bacterial antigen cross-reactive with a host protein (molecular mimicry), that then spreads over time to involve other targets. This implies that the greatest chance to selectively ablate an autoimmune condition is at the earliest point possible.

Serologic autoantibody biomarkers have even been used successfully to identify and treat presymptomatic high risk individuals, for example, with the anti-pan-T cell antibody tepelizumab in IDDM[103] and rituximab in RA.[104] Unlike these other disorders, the absence of any known serologic antibodies associated with MS has hampered an understanding of presymptomatic immune changes in this disease. Recently, however, a highly specific serum antibody biomarker for MS has been identified in a subset of presymptomatic individuals[105] and could be used—possibly with genetic or other markers—to design a presymptomatic trial in MS.

Importantly, clinical trials using intensive immunosuppression with broadly active agents, including cladribine[106] and alemtuzumab,[107] at early clinical timepoints in MS have shown impressive long-term results against late disability similar to AHSCT. In some patients elimination of OCB has been achieved. It is currently not known if purely B cell-targeted therapies initiated at clinical onset would be similarly effective, but one trial testing the anti-CD20 monoclonal antibody OCR in incident MS is now nearing completion.

Prospects for Primary Prevention

Exposure to EBV is a near-universal prerequisite for development of MS.[108] However, approximately 95% of the global population is infected with EBV at some point in their lives, making it impossible to use serologic evidence of EBV as a biomarker for MS risk. The recently identified MS-specific antibody present in presymptomatic serum samples appears to represent a unique immune response to the lytic EBV protein BRRF2 that somehow sets the stage for MS, possibly through molecular mimicry. In addition, antibodies against EBNA1 are increased in MS patients carrying the risk allele HLA-DRB1*15:01, which increases the likelihood of cross-reactivity to CNS autoantigens.[109–111] If an effective vaccine against EBV was developed and deployed population-wide, would MS disappear? One concern with this approach is that the vaccine itself could precipitate MS in some individuals via molecular mimicry. Alternatively, because the risk of MS following primary infection with EBV increases substantially when infection occurs later in childhood, resulting in a stronger anti-viral immune response and symptoms of infectious mononucleosis, a vaccine might be safe and protective against MS if administered early in life. Although EBV is by far the strongest microbial species associated with MS, other bacteria and viruses (eg, the commensal gut bacterium Akkermansia muciniphila)[112] has also been strongly associated. Therefore, elimination of any single infectious trigger might not be adequate.

Remyelination and Neural Repair

Another major unmet need in MS is reversal of existing disability. MS is often described as a demyelinating disease and many plaques naturally undergo remyelination. Remyelinated plaques, or shadow plaques, show less dense and shorter segments of myelination compared to normal white matter. However, not all plaques remyelinate and the observation of oligodendroglial precursor cells (OPCs) at plaque borders that are not remyelinating denuded axons suggest that enhancing myelin repair within lesions might be feasible. Despite multiple trials, to date no investigational product has demonstrated the capacity to reduce MS-related disability through myelin repair. Clemastine, a generic over-the-counter antihistamine that also works as an antagonist to the M1 muscarinic receptor, was identified by a high throughput screen of approved drugs capable of stimulating myelination.[113] In a placebo-controlled trial, clemastine reduced the visual evoked potential (VEP) latency in relapsing MS patients with prolonged VEP baseline latencies, a physiologic effect that is consistent with remyelination of the visual pathway.[114] Several studies of clemastine are ongoing, some in combination with metformin, as well as more selective M1 muscarinic receptor antagonists,[115] which may help aged OPCs respond to myelin repair signals (Table).

Although demyelination certainly occurs in MS plaques, tissue injury extends beyond myelin damage. Acute MS plaques have a high density of axonal transections, ultimately resulting in neuronal death from Wallerian degeneration. Demyelinated axons are at increased risk of transection and an increasing body of data indicates that remyelination of denuded axons can protect against subsequent neuronal loss.[116] Therefore, myelin repair might promote relapse recovery or even reverse deficits directly attributable to demyelination, and also prevent neuronal death by protecting the axon. The challenge with the disability endpoints that are traditionally used to assess effects of treatments for MS is that myelin repair may reverse disability only in patients with deficits because of failed remyelination. Deficits caused by neuronal death are irreversible and no amount of remyelination of a transected axon will prevent Wallerian degeneration. Therefore, selection of deficits known to be caused by demyelination, such as heat- or fatigue-related symptoms or paroxysmal deficits (eg, trigeminal neuralgia, glossopharyngeal neuralgia, and paroxysmal vertigo) may be better initial targets for myelin repair strategies than improvement in composite or global measures of MS disability. Last, because remyelination has the potential to protect demyelinated neurons from eventual transection, the impact of effective remyelination might not be reversal of existing disability, but protection against future irreversible neuronal loss and clinical decline. With this property in mind, remyelination studies might take years to complete to show an impact of remyelination on relevant endpoints. Such studies would also need to carefully select the optimal patient population to study, as the benefit of remyelination may be greater in more recently formed plaques. Reduction in thalamic atrophy in early relapsing MS patients could be an endpoint for a proof-of-concept study of a remyelinating therapy used in combination with a highly effective DMT that would prevent new lesion formation.

Defining Complete Remission and Cure

In cancer medicine, the definition of a complete remission requires disappearance of all signs of cancer. This does not always mean the cancer has been cured, but it is a prerequisite to cure. For example, in children with acute lymphocytic leukemia, a B cell malignancy, patients who maintain complete remission status while treatment-free for 4 years have less than a 1% chance of relapse; they are effectively cured.[117]

We could similarly consider criteria for complete remission in MS, defined as no evidence of disease activity (NEDA) by clinical and MRI lesion criteria, reducing age-adjusted MRI volume change to healthy levels, normalizing NfL and GFAP levels in serum, and eliminating OCBs and elevated Ig indices in CSF. As therapeutic approaches for MS become even more effective, the current model of lifetime treatment may be replaced with a model more akin to cancer protocols, consisting of highly effective induction therapy until complete remission is sustained, followed by monitoring off-treatment (or possibly with less-intensive maintenance therapy), and either periodic reinduction or discharge as a successful cure.

These are exciting prospects, but they rest on still insecure footing and many questions remain. Would all patients be treated similarly or will there be protocols differ based on presenting features? Is the normalization of CSF parameters tantamount to complete remission? What duration of remission off-treatment would be required to consider the patient cured? Given the spectacular progress to date, it may not be presumptuous to contemplate the possibility that MS could be the first chronic autoimmune disease to be cured.

One other point seems certain. The questions that require answers in the future are unlikely to be answered solely or even largely by the pharmaceutical industry, or by traditional billion-dollar trials. New technologies, such as novel platforms for immune diagnosis and monitoring, and artificial intelligence and machine learning, to interrogate large medical databases, among others, are certain to assume an increasing important role. Some questions—such as those that could lead to use of lower amounts of drug—cannot be reasonably asked of industry partners. Success will require development of closely interlocking teams and consortia bridging institutions and disciplines, including non-traditional ones. For those entering the field today, one thing seems certain: the work will be more interesting and rewarding than ever before and also likely to provide a few more surprises. Most important, MS will become—to paraphrase Einstein—a simpler problem to treat, prevent, and cure.

Author Contributions

J.J.S., B.A.C., and S.L.H. each contributed to the conception and design of the manuscript; J.J.S., B.A.C., and S.L.H. each contributed to the interpretation of the studies included in the manuscript; J.J.S., B.A.C., and S.L.H. each contributed to drafting the text and preparing the figures.

Potential Conflicts of Interest

J.J.S. has received research support from Novartis AG and Genentech and has received consulting fees from IgM Biosciences and TG Therapeutics. B.A.C. has received research support from Genentech and Kyverna and has received consulting fees from Alexion, Atara, Autobahn, Avotres, Biogen, Boston Pharma, EMD Serono, Hexal/Sandoz, Horizon, Kyverna, Neuron23, Novartis, Sanofi, Siemens and TG Therapeutics. S.L.H. has received consulting fees from BD, Gilead, Moderna, NGM Bio, Nurix Therapeutics, and Pheno Therapeutics, has received travel

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REFERENCE

1. Harding K, Williams O, Willis M, et al. Clinical outcomes of escalation vs early intensive disease-modifying therapy in patients with multiple sclerosis. JAMA Neurol 2019;76:536–541.
2. Brown JWL, Coles A, Horakova D, et al. Association of Initial Disease-Modifying Therapy with Later Conversion to secondary progressive multiple sclerosis. JAMA 2019;321:175–187.
3. He A, Merkel B, Brown JWL, et al. Timing of high-efficacy therapy for multiple sclerosis: a retrospective observational cohort study. Lancet Neurol 2020;19:307–316.
4. Buron MD, Chalmer TA, Sellebjerg F, et al. Initial high-efficacy disease-modifying therapy in multiple sclerosis. Neurology 2020;95: e1041–e1051.
5. Spelman T, Magyari M, Piehl F, et al. Treatment escalation vs immediate initiation of highly effective treatment for patients with relapsing–remitting multiple sclerosis: data from 2 different National Strategies. JAMA Neurol 2021;78:1197–1204.
6. Kalincik T, Sharmin S, Roos I, et al. Comparative effectiveness of autologous hematopoietic stem cell transplant vs Fingolimod, Natalizumab, and Ocrelizumab in highly active relapsing–remitting multiple sclerosis. JAMA Neurol 2023;80:702–713.
7. Cerqueira JJ, Berthele A, Cree BAC, et al. Long-term treatment with Ocrelizumab in patients with early-stage relapsing MS. Neurology 2025;104:e210142.
8. Hauser S, Waubant E, Arnold DL, et al. B-cell depletion with rituximab in relapsing–remitting multiple sclerosis. N Engl J Med 2008;358:676–688.
9. Hauser SL, Bar-Or A, Comi G, et al. Ocrelizumab versus interferon Beta-1a in relapsing multiple sclerosis. N Engl J Med 2017;376:221–234.
10. Cohen JA, Coles AJ, Arnold DL, et al. Alemtuzumab versus interferon beta 1a as first-line treatment for patients with relapsing–remitting multiple sclerosis: a randomised controlled phase 3 trial. Lancet 2012;380:1819–1828.
11. Coles AJ, Twyman CL, Arnold DL, et al. Alemtuzumab for patients with relapsing multiple sclerosis after disease-modifying therapy: a randomised controlled phase 3 trial. Lancet 2012;380:1829–1839.
12. Polman CH, O'Connor PW, Havrdova E, et al. A randomized, placebo-controlled trial of Natalizumab for relapsing multiple sclerosis. N Engl J Med 2006;354:899–910.
13. Rudick RA, Stuart WH, Calabresi PA, et al. Natalizumab plus interferon Beta-1a for relapsing multiple sclerosis. N Engl J Med 2006; 354:911–923.
14. Sabatino JJ, Pröbstel A-K, Zamvil SS. B cells in autoimmune and neurodegenerative central nervous system diseases. Nat Rev Neurosci 2019;20:728–745.
15. Lindner M, Klotz L, Wiendl H. Mechanisms underlying lesion development and lesion distribution in CNS autoimmunity. J Neurochem 2018;146:122–132.
16. Dobson R, Ramagopalan S, Davis A, Giovannoni G. Cerebrospinal fluid oligoclonal bands in multiple sclerosis and clinically isolated syndromes: a meta-analysis of prevalence, prognosis and effect of latitude. J Neurol Neurosurg Psychiatry 2013;84:909–914.
17. von Büdingen H-C, Kuo TC, Sirota M, et al. B cell exchange across the blood–brain barrier in multiple sclerosis. J Clin Invest 2012;122:24–28.
18. Obermeier B, Mentele R, Malotka J, et al. Matching of oligoclonal immunoglobulin transcriptomes and proteomes of cerebrospinal fluid in multiple sclerosis. Nat Med 2008;14:688–693.
19. Brändle SM, Obermeier B, Senel M, et al. Distinct oligoclonal band antibodies in multiple sclerosis recognize ubiquitous self-proteins. Proc Natl Acad Sci 2016;113:7864–7869.
20. Duddy M, Niino M, Adatia F, et al. Distinct effector cytokine profiles of memory and naive human B cell subsets and implication in multiple sclerosis. J Immunol 2007;178:6092–6099.
21. Bar-Or A, Fawaz L, Fan B, et al. Abnormal B-cell cytokine responses a trigger of T-cell-mediated disease in MS? Ann Neurol 2010;67:452–461.
22. Barr TA, Shen P, Brown S, et al. B cell depletion therapy ameliorates autoimmune disease through ablation of IL-6-producing B cells. J Exp Med 2012;209:1001–1010.
23. Li R, Rezk A, Miyazaki Y, et al. Proinflammatory GM-CSF-producing B cells in multiple sclerosis and B cell depletion therapy. Sci Transl Med 2015;7:310ra166.
24. Hauser SL, Kappos L, Arnold DL, et al. Five years of ocrelizumab in relapsing multiple sclerosis. Neurology 2020;95:e1854–e1867.
25. Wolinsky JS, Arnold DL, Brochet B, et al. Long-term follow-up from the ORATORIO trial of ocrelizumab for primary progressive multiple sclerosis: a post-hoc analysis from the ongoing open-label extension of the randomised, placebo-controlled, phase 3 trial. Lancet Neurol 2020;19:998–1009.
26. Cross AH, Stark JL, Lauber J, et al. Rituximab reduces B cells and T cells in cerebrospinal fluid of multiple sclerosis patients. J Neuroimmunol 2006;180:63–70.
27. Greenfield AL, Dandekar R, Ramesh A, et al. Longitudinally persistent cerebrospinal fluid B cells can resist treatment in multiple sclerosis. JCI Insight 2019;4:e126599. https://doi.org/10.1172/jci.insight.126599.
28. Graner M, Pointon T, Manton S, et al. Oligoclonal IgG antibodies in multiple sclerosis target patient-specific peptides. PLoS One 2020; 15:e0228883.
29. Paul F, Marignier R, Palace J, et al. International Delphi consensus on the management of AQP4-IgG+ NMOSD. Neurol Neuroimmunol Neuroinflamm 2023;10:e200124.
30. Guasp M, Dalmau J. Autoimmune Encephalitis. Med Clin North Am 2025;109:443–461.
31. Genç K, Dona DL, Reder AT. Increased CD80+ B cells in active multiple sclerosis and reversal by interferon β-1b therapy. J Clin Invest 1997;99:2664–2671.
32. Fraussen J, Claes N, Van Wijmeersch B, et al. B cells of multiple sclerosis patients induce autoreactive proinflammatory T cell responses. Clin Immunol 2016;173:1–9.
33. Harp CT, Ireland S, Davis LS, et al. Memory B cells from a subset of treatment-naïve relapsing–remitting multiple sclerosis patients elicit CD4+ T-cell proliferation and IFN-γ production in response to myelin basic protein and myelin oligodendrocyte glycoprotein. Eur J Immunol 2010;40:2942–2956.
34. Häusler D, Häusser-Kinzel S, Feldmann L, et al. Functional characterization of reappearing B cells after anti-CD20 treatment of CNS autoimmune disease. Proc Natl Acad Sci 2018;115:9773–9778.
35. Molnarfi N, Schulze-Topphoff U, Weber MS, et al. MHC class II-dependent B cell APC function is required for induction of CNS autoimmunity independent of myelin-specific antibodies. J Exp Med 2013;210:2921–2937.
36. Jelcic I, Nimer FA, Wang J, et al. Memory B cells activate brain-homing, autoreactive CD4+ T cells in multiple sclerosis. Cell 2018;175:85–100.
37. Wang J, Jelcic I, Mühlenbruch L, et al. HLA-DR15 molecules jointly shape an autoreactive T cell repertoire in multiple sclerosis. Cell 2020;183:1264–1281.e20.
38. Fan X, Jin T, Zhao S, et al. Circulating CCR7 + ICOS+ memory T follicular helper cells in patients with multiple sclerosis. PLoS One 2015;10:1–14.

39. Holm Hansen R, Talbot J, Højsgaard Chow H, et al. Increased intrathecal activity of follicular helper T cells in patients with relapsing–remitting multiple sclerosis. Neurol Neuroimmunol Neuroinflamm 2022;9:e200009.

40. Montalban X, Hauser SL, Kappos L, et al. Ocrelizumab versus placebo in primary progressive multiple sclerosis. N Engl J Med 2017; 376:209–220.

41. Kapoor R, Ho P-R, Campbell N, et al. Effect of natalizumab on disease progression in secondary progressive multiple sclerosis (ASCEND): a phase 3, randomised, double-blind, placebo-controlled trial with an open-label extension. Lancet Neurol 2018;17:405–415.

42. Horáková D, Boster A, Bertolotto A, et al. Proportion of alemtuzumab-treated patients converting from relapsing–remitting multiple sclerosis to secondary progressive multiple sclerosis over 6 years. Mult Scler J Exp Transl Clin 2020;6:2055217320972137.

43. Petereit HF, Rubbert-Roth A. Rituximab levels in cerebrospinal fluid of patients with neurological autoimmune disorders. Mult Scler 2009;15:189–192.

44. Sipe JC, Romine JS, Koziol JA, et al. Cladribine in treatment of chronic progressive multiple sclerosis. Lancet 1994;344:9–13. https://doi.org/10.1016/s0140-6736(94)91046-4.

45. Lublin F, Miller DH, Freedman MS, et al. Oral fingolimod in primary progressive multiple sclerosis (INFORMS): a phase 3, randomised, double-blind, placebo-controlled trial. Lancet 2016;387:1075–1084.

46. Cree BA, Arnold DL, Fox RJ, et al. Long-term efficacy and safety of siponimod in patients with secondary progressive multiple sclerosis: analysis of EXPAND core and extension data up to >5 years. Mult Scler 2022;28:1591–1605.

47. Henry RG, Shieh M, Okuda DT, et al. Regional grey matter atrophy in clinically isolated syndromes at presentation. J Neurol Neurosurg Psychiatry 2008;79:1236–1244.

48. Barkhof F. MRI in multiple sclerosis: correlation with expanded disability status scale (EDSS). Mult Scler 1999;5:283–286.

49. Hauser SL, Cree BAC. Treatment of multiple sclerosis: a review. Am J Med 2020;133:1380–1390.e2.

50. University of California SFM-ET, Cree BAC, Hollenbach JA, et al. Silent progression in disease activity–free relapsing multiple sclerosis. Ann Neurol 2019;85:653–666.

51. Kappos L, Wolinsky JS, Giovannoni G, et al. Contribution of relapse-independent progression vs relapse-associated worsening to overall confirmed disability accumulation in typical relapsing multiple sclerosis in a pooled analysis of 2 randomized clinical trials. JAMA Neurol 2020;77:1132–1140.

52. Lassmann H, van Horssen J, Mahad D. Progressive multiple sclerosis: pathology and pathogenesis. Nat Rev Neurol 2012;8:647–656.

53. Mahad DH, Trapp BD, Lassmann H. Pathological mechanisms in progressive multiple sclerosis. Lancet Neurol 2015;14:183–193.

54. Kutzelnigg A, Lucchinetti CF, Stadelmann C, et al. Cortical demyelination and diffuse white matter injury in multiple sclerosis. Brain 2005;128:2705–2712.

55. Lucchinetti CFC, Popescu BFGB, Bunyan RF, et al. Inflammatory cortical demyelination in early multiple sclerosis. N Engl J Med 2011;365:2188–2197.

56. Choi SR, Howell OW, Carassiti D, et al. Meningeal inflammation plays a role in the pathology of primary progressive multiple sclerosis. Brain 2012;135:2925–2937.

57. Serafini B, Rosicarelli B, Magliozzi R, et al. Detection of ectopic B-cell follicles with germinal centers in the meninges of patients with secondary progressive multiple sclerosis. Brain Pathol 2004;14: 164–174.

58. Frischer JM, Bramow S, Dal-bianco A, et al. The relation between inflammation and neurodegeneration in multiple sclerosis brains. Brain 2009;132:1175–1189.

59. Hayashi T, Morimoto C, Burks JS, et al. Dual-label immunocytochemistry of the active multiple sclerosis lesion: major histocompatibility complex and activation antigens. Ann Neurol 1988;24:523–531.

60. Machado-Santos J, Saji E, Tröscher AR, et al. The compartmentalized inflammatory response in the multiple sclerosis brain is composed of tissue-resident CD8+ T lymphocytes and B cells. Brain 2018;141:2066–2082.

61. Magliozzi R, Howell OW, Reeves C, et al. A gradient of neuronal loss and meningeal inflammation in multiple sclerosis. Ann Neurol 2010;68:477–493.

62. Androdias G, Reynolds R, Chanal M, et al. Meningeal T cells associate with diffuse axonal loss in multiple sclerosis spinal cords. Ann Neurol 2010;68:465–476.

63. Absinta M, Maric D, Gharagozloo M, et al. A lymphocyte–microglia–astrocyte axis in chronic active multiple sclerosis. Nature 2021;597:709–714.

64. van den Bosch AMR, van der Poel M, Fransen NL, et al. Profiling of microglia nodules in multiple sclerosis reveals propensity for lesion formation. Nat Commun 2024;15:1667.

65. Zrzavy T, Hametner S, Wimmer I, et al. Loss of 'homeostatic' microglia and patterns of their activation in active multiple sclerosis. Brain 2017;140:1900–1913.

66. Yong VW. Microglia in multiple sclerosis: protectors turn destroyers. Neuron 2022;110:3534–3548.

67. Grajchen E, Hendriks JJA, Bogie JFJ. The physiology of foamy phagocytes in multiple sclerosis. Acta Neuropathol Commun 2018; 6:124.

68. Lee H-G, Lee J-H, Flausino LE, Quintana FJ. Neuroinflammation: an astrocyte perspective. Sci Transl Med 2023;15:eadi7828.

69. Franklin RJM, ffrench-Constant C. Remyelination in the CNS: from biology to therapy. Nat Rev Neurosci 2008;9:839–855.

70. Hauser SL, Kappos L, Montalban X, et al. Safety of Ocrelizumab in patients with relapsing and primary progressive multiple sclerosis. Neurology 2021;97:e1546–e1559.

71. Bischof A, Papinutto N, Keshavan A, et al. Spinal cord atrophy predicts progressive disease in relapsing multiple sclerosis. Ann Neurol 2022;91:268–281.

72. Abdelhak A, Antweiler K, Kowarik MC, et al. Serum glial fibrillary acidic protein and disability progression in progressive multiple sclerosis. Ann Clin Transl Neurol 2024;11:477–485.

73. Laaksonen S, Saraste M, Sucksdorff M, et al. Early prognosticators of later TSPO-PET-measurable microglial activation in multiple sclerosis. Mult Scler Relat Disord 2023;75:104755.

74. Singhal T, O'Connor K, Dubey S, et al. Gray matter microglial activation in relapsing vs progressive MS. Neurol Neuroimmunol Neuroinflamm 2019;6:e587.

75. Sucksdorff M, Matilainen M, Tuisku J, et al. Brain TSPO-PET predicts later disease progression independent of relapses in multiple sclerosis. Brain 2020;143:3318–3330.

76. Nutma E, Fancy N, Weinert M, et al. Translocator protein is a marker of activated microglia in rodent models but not human neurodegenerative diseases. Nat Commun 2023;14:5247.

77. Hauser SL, Giovannoni G, Filippi M, et al. The patient impact of 11 years of ocrelizumab treatment in multiple sclerosis: long-term data from the phase III OPERA and ORATORIO studies, 2024.

78. Hauser SL, Bar-Or A, Weber MS, et al. Association of higher Ocrelizumab exposure with reduced disability progression in multiple sclerosis. Neurol Neuroimmunol Neuroinflamm 2023;10:e200094.

79. Cross A, Hauser S, Wiendl H, et al. B-cell depletion and efficacy outcomes of Ofatumumab are consistent across different body mass index categories: insights from ASCLEPIOS I and II trials (P9-6.002). Neurology 2024;102:5874.

80. Lassmann H. Pathogenic mechanisms associated with different clinical courses of multiple sclerosis. Front Immunol 2019;9:3116. https://doi.org/10.3389/fimu.2018.03116.

81. Cree BAC, Bennett JL, Kim HJ, et al. Inebilizumab for the treatment of neuromyelitis optica spectrum disorder (N-MOmentum): a double-blind, randomised placebo-controlled phase 2/3 trial. The Lancet 2019;394:1352–1363.

82. Agius MA, Klodowska-Duda G, Maciejowski M, et al. Safety and tolerability of inebilizumab (MEDI-551), an anti-CD19 monoclonal antibody, in patients with relapsing forms of multiple sclerosis: results from a phase 1 randomised, placebo-controlled, escalating intravenous and subcutaneous dose study. Mult Scler 2019;25:235–245.

83. Mitra A, Barua A, Huang L, et al. From bench to bedside: the history and progress of CAR T cell therapy. Front Immunol 2023;14: 1188049. https://doi.org/10.3389/fimmu.2023.1188049.

84. Müller F, Taubmann J, Bucci L, et al. CD19 CAR T-cell therapy in autoimmune disease — a case series with follow-up. N Engl J Med 2024;390:687–700. https://doi.org/10.1056/NEJMoa2308917.

85. Fischbach F, Richter J, Pfeffer LK, et al. CD19-targeted chimeric antigen receptor T cell therapy in two patients with multiple sclerosis. Med 2024;5:550–558.e2.

86. Wang W, He S, Zhang W, et al. BCMA-CD19 compound CAR T cells for systemic lupus erythematosus: a phase 1 open-label clinical trial. Ann Rheum Dis 2024;83:1304–1314.

87. Bodansky A, Yu DJL, Rallistan A, et al. Unveiling the proteome-wide autoreactome enables enhanced evaluation of emerging CAR T cell therapies in autoimmunity. J Clin Invest 2024;134:e180012. https://doi.org/10.1172/JCI180012.

88. Ellithi M, Elsallab M, Lunning MA, et al. Neurotoxicity and rare adverse events in BCMA-directed CAR T cell therapy: a comprehensive analysis of real-world FAERS data. Transplant Cell Ther 2025;31:71.e1–71.e14.

89. Simic MS, Watchmaker PB, Gupta S, et al. Programming tissue-sensing T cells that deliver therapies to the brain. Science 2024; 386:eadl4237.

90. Bui TA, Mei H, Sang R, et al. Advancements and challenges in developing in vivo CAR T cell therapies for cancer treatment. EBioMedicine 2024;106:105266.

91. Autologous stem-cell transplantation for severe scleroderma. N Engl J Med 2018;378:1066–1067.

92. Muraro PA, Mariottini A, Greco R, et al. Autologous haematopoietic stem cell transplantation for treatment of multiple sclerosis and neuromyelitis optica spectrum disorder — recommendations from ECTRIMS and the EBMT. Nat Rev Neurol 2025;21:140–158. https://doi.org/10.1038/s41582-024-01050-x.

93. Ruder J, Docampo MJ, Rex J, et al. Dynamics of T cell repertoire renewal following autologous hematopoietic stem cell transplantation in multiple sclerosis. Sci Transl Med 2022;14:eabq1693.

94. Larsson D, Åkerfeldt T, Carlson K, Burman J. Intrathecal immunoglobulins and neurofilament light after autologous haematopoietic stem cell transplantation for multiple sclerosis. Mult Scler 2020;26: 1351–1359.

95. Bruton OC. Agammaglobulinemia. Pediatrics 1952;9:722–728. https://doi.org/10.1542/peds.9.6.722.

96. Smith C, Berglöf A. In: Adam MP, Feldman J, Mirzaa GM, et al., eds. *X-Linked Agammaglobulinemia [Internet]*. GeneReviews®, 2001 (Seattle, WA). Available at: https://www.ncbi.nlm.nih.gov/books/NBK1453/.

97. Montalban X, Vermersch P, Arnold DL, et al. Safety and efficacy of evobrutinib in relapsing multiple sclerosis (evolutionRMS1 and evolutionRMS2): two multicentre, randomised, double-blind, active-controlled, phase 3 trials. Lancet Neurol 2024;23:1119–1132.

98. Oh J, Arnold DL, Cree BAC, et al. 154. Efficacy and safety of Tolebrutinib versus Teriflunomide in relapsing multiple sclerosis: results from the phase 3 GEMINI 1 and 2 trials. Mult Scler Relat Disord 2024;92:106115.

99. Fox RJ, Bar-Or A, Traboulsee A, et al. 153. Efficacy and safety of Tolebrutinib versus placebo in non-relapsing secondary progressive multiple sclerosis: results from the phase 3 HERCULES trial. Mult Scler Relat Disord 2024;92:106114.

100. Petersen MA, Ryu JK, Akassoglou K. Fibrinogen in neurological diseases: mechanisms, imaging and therapeutics. Nat Rev Neurosci 2018;19:283–301.

101. Babbe H, Roers a, Waisman a, et al. Clonal expansions of CD8(+) T cells dominate the T cell infiltrate in active multiple sclerosis lesions as shown by micromanipulation and single cell polymerase chain reaction. J Exp Med 2000;192:393–404.

102. Vehik K, Bonifacio E, Lernmark Å, et al. Hierarchical order of distinct autoantibody spreading and progression to type 1 diabetes in the TEDDY study. Diabetes Care 2020;43:2066–2073.

103. Herold KC, Bundy BN, Long SA, et al. An anti-CD3 antibody, Teplizumab, in relatives at risk for type 1 diabetes. N Engl J Med 2019;381:603–613. https://doi.org/10.1056/NEJMoa1902226.

104. Frazzei G, Cramer SHM, Landewé RBM, et al. The effect of rituximab on patient reported outcomes in the preclinical phase of rheumatoid arthritis: 2 year data from the PRAIRI study. RMD Open 2024;10:e004622.

105. Zamecnik CR, Sowa GM, Abdelhak A, et al. An autoantibody signature predictive for multiple sclerosis. Nat Med 2024;30:1300–1308.

106. Rejdak K, Stelmasiak Z, Grieb P. Cladribine induces long lasting oligoclonal bands disappearance in relapsing multiple sclerosis patients: 10-year observational study. Mult Scler Relat Disord 2019;27:117–120.

107. Möhn N, Pfeuffer S, Ruck T, et al. Alemtuzumab therapy changes immunoglobulin levels in peripheral blood and CSF. Neurol Neuroimmunol Neuroinflamm 2019;7:e654.

108. Bjornevik K, Münz C, Cohen JI, Ascherio A. Epstein–Barr virus as a leading cause of multiple sclerosis: mechanisms and implications. Nat Rev Neurol 2023;19:160–171. https://doi.org/10.1038/s41582-023-00775-5.

109. Tengvall K, Huang J, Hellström C, et al. Molecular mimicry between Anoctamin 2 and Epstein–Barr virus nuclear antigen 1 associates with multiple sclerosis risk. Proc Natl Acad Sci 2019; 116:16955–16960.

110. Thomas OG, Rickinson A, Palendira U. Epstein–Barr virus and multiple sclerosis: moving from questions of association to questions of mechanism. Clin Transl Immunol 2023;12:e1451.

111. Ganta KK, McManus M, Blanc R, et al. Acute infectious mononucleosis generates persistent, functional EBNA-1 antibodies with high cross-reactivity to alpha-crystalline beta. Cell Rep 2025;44:115709.

112. Zhou X, Baumann R, Gao X, et al. Gut microbiome of multiple sclerosis patients and paired household healthy controls reveal associations with disease risk and course. Cell 2022;185:3467–3486.e16.

113. Mei F, Fancy SPJ, Shen Y-AA, et al. Micropillar arrays as a high-throughput screening platform for therapeutics in multiple sclerosis. Nat Med 2014;20:954–960.

114. Green AJ, Gelfand JM, Cree BA, et al. Clemastine fumarate as a remyelinating therapy for multiple sclerosis (ReBUILD): a randomised, controlled, double-blind, crossover trial. The Lancet 2017;390: 2481–2489.

115. Poon MM, Lorrain KI, Stebbins KJ, et al. Targeting the muscarinic M1 receptor with a selective, brain-penetrant antagonist to promote remyelination in multiple sclerosis. Proc Natl Acad Sci 2024; 121:e2407974121.

116. Irvine KA, Blakemore WF. Remyelination protects axons from demyelination-associated axon degeneration. Brain 2008;131:1464–1477.

117. Pui CH, Pei D, Campana D, et al. A revised definition for cure of childhood acute lymphoblastic leukemia. Leukemia 2014;28:2336–2343.